

DSC 670 Term Project Instructions

Throughout the term, you will be working on a project that has four milestones, which are due in Weeks 2, 4, 8, and 11. The goal of the project is to build on the materials covered during the course. You'll be creating an application that uses Gen AI, which means the milestones are application driven, which may be different than other projects. I'll provide detailed examples that you can refer to throughout the course.

Milestone 1 – Pick Your Project

Milestone 1 is due in Week 2.

At this early a stage, it can be a pretty daunting task to pick a project. However, given the use cases you've seen so far, you can likely think of something that would be done more efficiently and effectively with generative AI. For this milestone, consider one or more ideas, do some basic research (e.g., a Google search on your idea to see if it's been done before), and articulate your idea. It's understood your idea and project may change as you learn more about generative AI.

For this first milestone, you need to create a formal paper with the following sections:

- A description of your idea
- Your main goal(s)
- Your *initial* thoughts about how you might approach the problem (you'll almost certainly change this as you learn more).

Your Milestone 1 paper **must** follow [this format provided in the APA Help Center](#).

Milestone 2 – Refine Your Project

Milestone 2 is due in Week 4.

Now that you've seen more types of GAI models, refine your project based on what you've learned. For this milestone, you'll modify what you did for Milestone 1 and add your initial design including the following details:

- Any updates to your problem statement.
- The type of generative AI you're going to use in the solution.
- At least five prompt experiments to see if your problem has the potential to be solved using the model you chose.
- The way, generally, you'd like to fine-tune the model. You may have to look ahead to Week 7 for more info on this.

You are expected to submit a formal paper and any code (if you're using an API) you created. Your paper must include your prompts, although you can optionally have your prompts in your code. Your Milestone 2 paper **must** follow [this format provided in the APA Help Center](#).

Milestone 3 – Build Your First Model

Milestone 3 is due in Week 8.

Now that you have a full picture of how to fine-tune GAI, build your first model using OpenAI. You may have to pick a dataset if you haven't already, but building the model using OpenAI should be pretty straightforward after Week 7.

For this milestone:

- Create the model. (You will fully fine-tune – don't use LoRA.)
- Track your model build process using the endpoints shown in Chapter 7.
- Document the metrics as shown in Chapter 7.

Build the milestone using a Jupyter Notebook (no other Python is acceptable) and include Markdown that explains your thought process and any commentary about how things went (e.g., did the fine-tuning work?). I'm looking for your commentary and evaluation in your own words.

Milestone 4 – Create Your Application to Finish the Project!

Milestone 4 is due in Week 11.

Use Streamlit to create an application that uses your solution. This deliverable is where you polish up what you've been doing and surface it in a web application. It's entirely possible you've already either started or even completed this web app in an earlier Milestone.

Submit your code as one or more .py files. **Please also submit a short presentation (PowerPoint or Sway) describing your project, along with screen shots of your running application.** Optionally, you can create a video that does the same.

Remember – your GitHub repository can act as a portfolio for potential employers! I would highly suggest using this to submit your work, so you can fill it with good content that demonstrates the projects you are working on! I also suggest that you keep your GitHub repository private to protect your work from being used by other people online.